LECTURE ACTIVITY 15: LINGERING EFFECTS OF SLEEP DEPRIVATION – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is heavily adapted from one created by the STAT 1110 Course Coordination Team, which in turn is modified from "Introduction to Statistical Investigations," by Tintle et. al.

Learning Objectives:

By the end of this activity, you should be able to

1) State the null and the alternative hypotheses in terms of "no association" versus "there is an association" as well as in terms of comparing long-run means for two categories of the explanatory variable (i.e., μ_1 and μ_2).
2) When comparing two quantitative distributions, identify which has the larger mean, median, and standard deviation.
3) Identify whether there is likely an association between a binary categorical variable and a quantitative response variable.
4) Implement the 3S strategy: find a statistic, simulate, and compute the strength of evidence against observed study results happening by chance alone.
5) Describe how to use cards to simulate what outcomes (in terms of difference in sample means) are to be expected in repeated random assignments, if there is no association between the two variables.
6) Conduct a simulation-based analysis when evaluating claims about two long-run means.
7) Find, interpret, and/or evaluate the p-value for a test of two long-run means.
8) Use the 2SD method to find and interpret a 95% confidence interval for the difference in long-run means.
9) State a complete conclusion about the alternative hypothesis (and null hypothesis) based on the magnitude of the p-value and the study design.

Step 1: Formulate a Research Question

Many students pull "all-nighters" when they have an important exam or a pressing assignment. Concerns that may arise include: Can you really function well the next day after a sleepless night? What about several days later: Can you recover from a sleepless night by getting a full night's sleep on the following nights?

Question 1

What is your research conjecture about whether or not one can recover from a sleepless night by getting a full night's sleep on the following nights? Explain.

Solution: Answers will vary. I conjecture that one full night of sleep may not completely erase the effects of a sleepless night. Sleep deprivation could interfere with consolidation of newly learned information, so performance may remain worse even after two recovery nights.

Step 2: Design a Study and Collect Data

A study in 2000[1] explored the delayed effects of sleep deprivation on learning; in particular, by exploring how well the subjects performed a visual discrimination task.

First, subjects were shown a "target screen," consisting of a grid of straight line segments and one "fixation point" demarcated by either a "T" (as in image (a) of Figure (1) below) or an "L" (as in image (a) of Figure (1) below), for 15 milliseconds. Subjects were then shown a blank screen, followed by a "masking screen" (like image (c) of Figure () below). Finally, subjects were asked two questions about the target screen: whether the fixation point had an "L" or a "T," and whether the lines slanted horizontally or vertically.

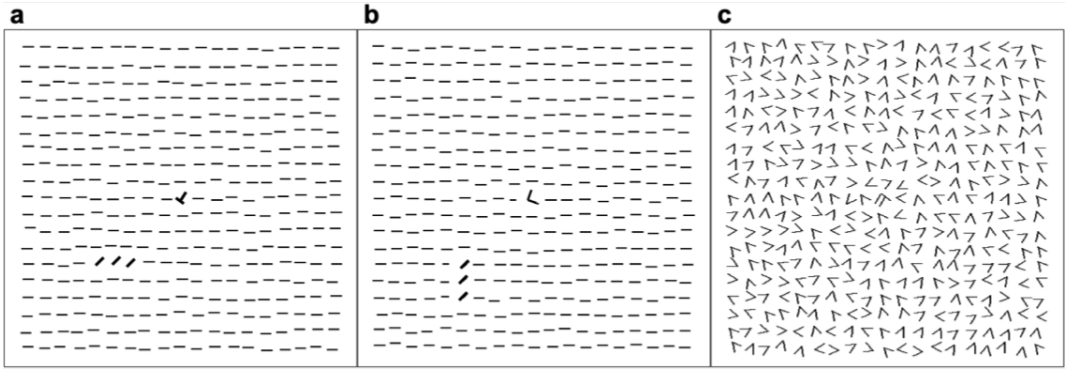


Figure 1: Target screens (a) and (b), and masking screen (c). Supplemental figure from the original paper; all rights retained by the original authors.

Twenty-one volunteers, aged 18 to 25 years, were first trained on on the visual discrimination task. After the training period, subjects were tested. Performance was recorded as the minimum time (in milliseconds) between the appearance of stimuli and an accurate response. Following these baseline measurements, one group was randomly assigned to be deprived of sleep for 30 hours, followed by two full nights of unrestricted sleep, whereas the other group was allowed to get unrestricted sleep on all three nights. Following this, both groups were retested on the task to see how well they remembered the training from the first day. Researchers recorded the improvement in performance as the decrease in time required at retest compared to training.

For example, if someone took 5 milliseconds (ms) to respond at the beginning of the study and then 2 ms, to respond at the end, the improvement score is 3 ms. But if someone took 2 ms at the beginning and then 5 ms at the end, the improvement score is -3 ms.

The goal of the study was to see whether the improvement scores tend to be higher for the unrestricted sleep treatment than for the sleep deprivation treatment.

[1] Stickgold, R., James, L. & Hobson, J. Visual discrimination learning requires sleep after training. *Nat Neurosci* 3, 1237–1238 (2000). https://doi.org/10.1038/81756

Question 2

Identify the two variables in this study. Then, classify each of them as either categorical or quantitative, and identify which variable is the explanatory variable and which is the response variable.

- **Variable 1:** Sleep Condition (unrestricted sleep or sleep deprived)
  - Classification: √ **Categorical** ◯ Quantitative
  - Type: √ **Explanatory** ◯ Response

- **Variable 2:** Improvement score (milliseconds)
  - Classification: ◯ Categorical √ **Quantitative**
  - Type: ◯ Explanatory √ **Response**

Question 3

Was this a randomized experiment, experiment, or an observational study? Explain how you are deciding.

Solution: This was a randomized experiment because the researchers randomly assigned the volunteers to either unrestricted sleep or sleep deprivation.

We'll think of one observational unit as a trial of a subject carrying out the experiment once, so that the two **long-run means** can be defined as

- $\mu_{\text{unrestricted}}$: long-run mean improvement score for those who would get unrestricted sleep
- μ_{deprived}: long-run mean improvement score for those who would be sleep deprived.

Question 4

State the null and the alternative hypotheses in words and in symbols, to investigate whether or not sleep deprivation has a negative effect on improvement in performance on visual discrimination tasks.

Solution:

- Null hypothesis: the long-run average improvement score for those with unrestricted sleep is the same as the long-run average improvement score for those who are sleep-deprived
  - $H_0 : \mu_{\text{unrestricted}} = \mu_{\text{deprived}}$ or, equivalently, $H_0 : \mu_{\text{unrestricted}} - \mu_{\text{deprived}} = 0$
- Alternative hypothesis: the long-run average improvement score for those with unrestricted sleep is LARGER/HIGHER than the long-run average improvement score for those who are sleep-deprived
  - $H_0 : \mu_{\text{unrestricted}} > \mu_{\text{deprived}}$ or, equivalently, $H_0 : \mu_{\text{unrestricted}} - \mu_{\text{deprived}} > 0$

Step 3: Explore the Data

Here's the data, with positive values indicating better performance at retest than at training, and negative values indicating worse performance at retest than at training:

- **Unrestricted-sleep group's improvement scores (milliseconds)**:

  25.2, 14.5, -7.0, 12.6, 34.5, 45.6, 11.6, 18.6, 12.1, 30.5

- **Sleep-deprived group's improvement scores (milliseconds)**:

  -10.7, 4.5, 2.2, 21.3, -14.7, -10.7, 9.6, 2.4, 21.8, 7.2, 10.0

To look at graphical and numerical summaries of the data from the study, we'll use the go to the Multiple Means applet (you'll need to click on the applet titled "Multiple Means"). The sleep deprivation data have already been entered into the applet. Notice that the applet creates parallel dotplots, one for each study group. To add **box plots** to the plot, click on the radio box next to "Add boxplots."

Question 5

(a) Based on these graphs alone, which group (unrestricted or deprived) appears to have had the higher mean improvement? How are you deciding?

Solution: The unrestricted-sleep group appears to have the higher mean improvement because its dotplot/boxplot is shifted farther to the right and contains more large positive improvement scores.

(b) Based on the graphs alone, which group (unrestricted or deprived) appears to have had more variability in improvement? How are you deciding?

Solution: The unrestricted-sleep group appears to have slightly more variability because its values are more spread out around the mean and they cover a wider range of values.

(c) Notice also that the applet also computes numerical summaries of the data, such as the mean and standard deviation (SD) for the improvements in each group. Fill out the table below:

Unrestricted Group	Deprived Group
$n_{\text{unrestricted}} =$ 10	$n_{\text{deprived}} =$ 11
$\bar{x}_{\text{unrestricted}} =$ 19.82 ms	$\bar{x}_{\text{deprived}} =$ 3.90 ms
$s_{\text{unrestricted}} =$ 12.17 ms	$s_{\text{deprived}} =$ 14.73 ms

Table 1: Summary Statistics

Recall from earlier in the course that the standard deviation is a measure of variability. Relatively speaking, smaller standard deviation values indicate less variability and a distribution whose data values tend to cluster more closely together, compared to a distribution with a larger standard deviation.

Question 6

(a) Based on the numerical summaries reported in the Question 5(c) above, which group (unrestricted or deprived) had the higher mean improvement?

Solution: The unrestricted-sleep group had the higher sample mean improvement: 19.82 ms compared to 3.90 ms.

(b) Based on the numerical summaries reported in the Question 5(c) above, which group (unrestricted or deprived) had the higher variability in improvement?

Solution: The unrestricted-sleep group had the greater variability because its sample standard deviation was larger: 14.73 ms versus 12.17 ms.

Notice that the applet also reports the observed difference in sample means for the improvement scores of the two groups. This will be our statistic value for the study.

Question 7

Report the value of the statistic. Make note of the order of subtraction!

$$\bar{x}_{\text{unrestricted}} - \bar{x}_{\text{deprived}} = \underline{\quad 15.920 \text{ ms} \quad}$$

Without any extensive statistical analysis, does this difference in sample means strike you as a meaningful difference? Explain your answer.

Solution: Yes. A difference of 15.92 ms appears practically meaningful because the unrestricted group improved by about 16 ms more on average. However, inference is still needed to determine whether a difference this large could plausibly arise from random assignment alone.

⚠ Caution

Don't forget - measurement units are especially important for quantitative variables!

Step 4: Draw Inferences Beyond the Data

There are two possible explanations for why we observed the two groups to have different sample means for improvement in performance: (1) it occurred just by chance or (2) there is really an effect of sleep deprivation.

Think about how you might go about deciding whether the observed difference between the two sample means is statistically significant. Once again, the key question is how often random assignment alone would produce a difference in the groups at least as extreme as the difference observed in this study, if there really were no effect of sleep condition on improvement score. We addressed similar questions in the previous lecture activity ("Tagging Penguins"); the only change is that now the response variable is *quantitative* rather than *categorical*, so the relevant statistic is the difference in group *means* rather than the difference in group *proportions*. Also, once again, we use simulation to investigate how often such an

extreme difference would occur by chance (random assignment) alone (if the null hypothesis of no difference / no effect / no association were true). In other words, we will again employ the **3S Strategy**.

Statistic

We already have a natural statistic value: what we calculated in Question 7 above. As a reminder, this value represents the difference in the sample mean improvement scores between the two groups (unrestricted vs. deprived).

Simulation

Similar to the description of the simulation in Lecture Activity 14 ("Tagging Penguins"), we'll use index cards to conduct our simulation. Also like last time, we'll start with a tactile implementation of the simulation before moving to the applet.

First, I think it may be useful to run through the logic of the simulation one more time. If the null hypothesis is true, then sleep status has no effect on the improvement scores. If this is the case, it logically follows that each subject would have had roughly the same improvement score *regardless of which group they were assigned to*. So, our simulation will imagine repeatedly assigning each of the subjects to potentially different groups, and checking in how many of these "shuffles" (simulated samples) do we observe a statistic value as or more extreme as the one we saw in Question 7 above.

Question 8

(a) How many index cards do we need to conduct this simulation? ____**21**____

Solution: We'll need 21 cards, one for each subject.

(b) How would the cards be differentiated? In other words, what will we write on each index card?

Solution: On each card, we'll write one of the 21 observed improvement scores. The cards represent the response values.

To conduct one *simulated sample*, we should

1) shuffle the stack of cards well

2) randomly distribute cards into two stacks: one stack with 10 cards (the unrestricted group) and one with 11 (the sleep deprived group)

3) compute the simulated mean for each group and calculate the difference in simulated sample means.

Question 9

Carry out the simulation at least once. Then, report the following quantities:

- Re-randomized UNRESTRICTED group's mean: _____
- Re-randomized DEPRIVED group's mean: _____
- DIFFERENCE in sample means (unrestricted − deprived): _____

Add your statistic value (the difference in sample means) to our collective dotplot on the whiteboard.

Question 10

Just to make sure you understand: is your answer to the previous question the result of a simulated sample or the result of what happened in the original study?

√ **Result of a simulated sample**

◯ Result of what happened in the original study

As before with simulation-based analyses, we'd like more than just a handful of repetitions - we'd like thousands (if not tens of thousands!) to determine what is typical and what is not for the difference in group means, assuming that sleep condition has no impact on improvement score. Indeed, the Multiple Means applet helps us to exactly that! Navigate back to the applet, and:

1) Check the **Show Shuffle Options** box

2) Select the **Plot** display

3) Press **Shuffle Responses**.

Question 11

Describe what the applet is doing and how this relates to your null hypothesis from Question 4 above.

Solution: The applet keeps all 21 observed improvement scores fixed, shuffles the sleep-condition labels, and recomputes the difference in sample means. This models the null hypothesis because, if sleep condition has no effect, the observed scores could be assigned to either group without changing their meaning.

Now, to see many more possible values of the difference in sample means, assuming sleep condition has no impact on improvement, do the following in the Multiple Means applet:

1) Change Number of Shuffles from 1 to 9,999.

2) Press **Shuffle Responses**, to produce a total of 10,000 shuffles and re-randomized statistics.

Consider the null distribution showing the 10,000 could-have-been values of difference in sample means, assuming that sleep condition has no effect on improvement.

Question 12

What does one dot on the plot represent in the **context** of the study?

Solution: One dot represents one random reassignment of the 21 improvement scores into groups of 10 and 11, followed by calculation of the simulated difference in means, unrestricted minus deprived.

Question 13

Why does it make sense that this **null distribution** is centered around 0?

Solution: The null distribution is centered near 0 because the null hypothesis states that the two long-run means are equal. Across many random reassignments, positive and negative differences balance, so the average simulated difference is approximately zero.

Question 14

Where does the observed difference in sample means fall in this histogram: near the middle or out in a tail? Are there a lot of observations that are even more extreme than the observed difference, assuming sleep condition has no impact on improvement? How are you deciding?

Solution: The observed difference of 15.92 ms falls far in the right tail. Only a small fraction of shuffled differences are at least this large, so it is unusual under the null hypothesis.

Strength of Evidence

Question 15

Use the applet to find and report the p-value. To show your work, include a screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) null distribution, (2) statistic and (3) p-value. Tip: To find the p-value you should use the results of the original study (Question 7) and NOT the results from your first simulated sample (Question 9).

- My approximate simulation-based *p*-value is **0.005** **(answers may vary)**

Here is my screenshot:

Solution:

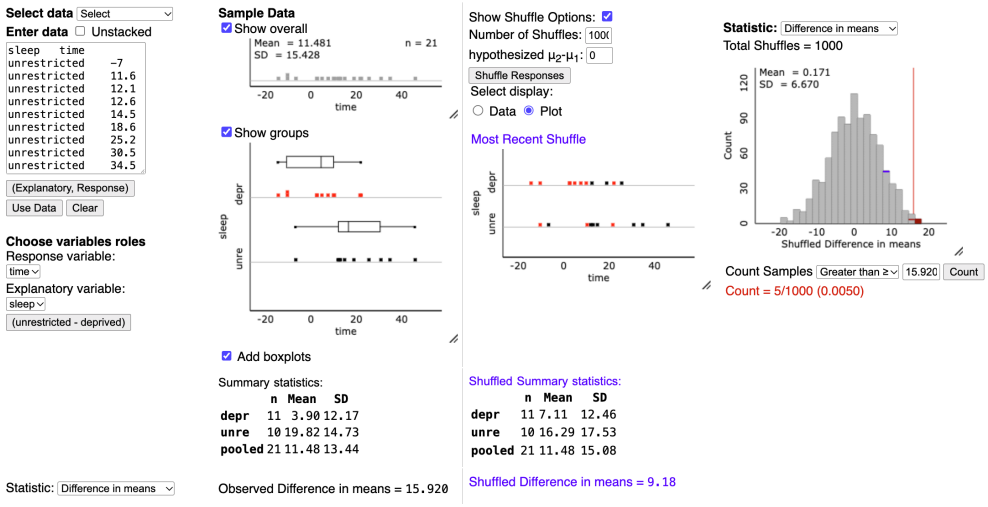


Figure 2: Screenshot from the Multiple Means Applet

Question 16

(a) ***Interpret*** this *p*-value using the **context** of the study.

Solution: The probability of obtaining a difference in sample means of 15.92 ms or larger merely from random assignment is about 0.005, if we assume that sleep condition truly had no effect on improvement score.

(b) ***Evaluate*** this *p*-value. How much evidence do the sample data provide against the null hypothesis? Choose one

- ◯ Not much evidence against the null hypothesis
- ◯ Moderate evidence against the null hypothesis
- ◯ Strong evidence against the null hypothesis
- √ **Very strong evidence against the null hypothesis**

Step 5: Formulate Conclusions

Question 17

Based on your analysis, give an overall conclusion in context. Make sure you refer to the parameter of interest in your conclusion.

Solution: There is very strong evidence that the long-run mean improvement score is higher for people who would receive unrestricted sleep than for people who would be sleep deprived.

Question 18

Generalization: Describe the population to which you would feel comfortable generalizing the results of this study. Be as specific as possible. **Tip**: remember that you can still make a generalization even if random sampling was not used in the study. But your generalization only applies to a very specific population.

Solution: Because the participants were volunteers rather than a random sample, generalization should be limited to 18- to 25-year-old people similar to these volunteers who would be willing and eligible to participate in a study like this one.

Question 19

Causation: Were the participants in the study randomly assigned to a sleep condition? How does this affect the scope of conclusion that you can draw?

Solution: Yes, the participants were randomly assigned to sleep condition. Random assignment supports a cause-and-effect conclusion: for people similar to those studied, the sleep-deprivation treatment caused lower improvement scores, on average.

Question 20

Estimation: Use the 2SD Method to approximate a 95% confidence interval for the parameter of interest which is the difference in long-run mean improvement score between subjects who would get unrestricted sleep and those who would be sleep deprived $(\mu_{\text{unrestricted}} - \mu_{\text{deprived}})$

Solution: $(15.920) \pm 2 \cdot \text{SD}_{\text{NULL}} = (15.920) \pm 2 \cdot 6.670 =$ **(2.58, 29.26)**

Question 21

Which of the following is the best interpretation of the confidence interval?

- ◯ I am 95% confident that the sample mean improvement score for people who would get unrestricted sleep is anywhere from 2 to 30ms smaller than the sample mean improvement score for people who would be sleep deprived, for 18 – 25-year-old people similar to the ones in the study who would be willing to volunteer to a study like this one.
- ◯ I am 95% confident that the sample mean improvement score for people who would get unrestricted sleep is anywhere from 2 to 30ms greater than the sample mean improvement score for people who would be sleep deprived, for 18 – 25-year-old people similar to the ones in the study who would be willing to volunteer to a study like this one.
- ◯ I am 95% confident that the long-run mean improvement score for people who would get unrestricted sleep is anywhere from 2 to 30ms smaller than the long-run mean improvement score for people who would be sleep deprived, for 18 – 25-year-old people similar to the ones in the study who would be willing to volunteer to a study like this one.
- √ **I am 95% confident that the long-run mean improvement score for people who would get unrestricted sleep is anywhere from 2 to 30ms greater than the long-run mean improvement score for people who would be sleep deprived, for 18 – 25-year-old**

people similar to the ones in the study who would be willing to volunteer to a study like this one.

Step 6: Look Back and Ahead

Question 22

Critique this study.

Solution: Strengths include random assignment, a control condition, and a clearly defined quantitative outcome. Limitations include the very small sample of 21 volunteers, restricted age range, and use of a single specialized visual task. The study supports causation for similar participants, but broader generalization requires a more representative sample and replication with other learning tasks and sleep-loss patterns.

Optional: Take It Further

If you feel adventurous or find these topics comfortable, explore the optional questions below. They are designed to stretch your thinking a bit further.

Using Medians Instead of Means

Could we have chosen a statistic other than the difference in group means to summarize how different the two groups' improvement scores were? Yes! For example we could have used the difference in group **medians**. Why might we do this? For one reason, the median is less affected by outliers than the mean (see Section 3.2).

To analyze the difference in group medians, we can carry out the 3S Strategy as before, except:

- We calculate the observed value of the difference in *medians* as the statistic.
- After we conduct the re-randomizing, we calculate the difference in medians for the re-randomized data. Then we repeat this process a large number of times.
- We determine the *p*-value by counting how many of the simulated statistics are at least as large as the observed value of the difference in medians.

Let's go back to the original study. Return to the Multiple Means applet and use the **Statistic** pull-down menu (on the left) to select **Difference in Medians**

1. From the Summary Statistics for the original data, record the median improvement score for each group.

Solution: The unrestricted-sleep median is 16.55 ms, the sleep-deprived median is 4.50 ms

- Unrestricted median: **16.55 ms**
- Deprived median: **4.50 ms**

2. What is the statistic that should be used for this new analysis?

Solution: The statistic is the difference in sample medians: $\text{median}_{\text{unrestricted}} - \text{median}_{\text{deprived}} = 16.55 - 4.50 = \mathbf{12.05\ ms.}$

3. Interpret the statistic in context.

Solution: The sample median improvement in the unrestricted-sleep group was 12.05 ms greater than the sample median improvement in the sleep-deprived group.

- Enter 10,000 for Number of Shuffles, and press Shuffle Responses.

4. What are the similarities between the null distribution you just found and the null distribution you obtained when using the difference in *means*?

Solution: Both null distributions are centered near 0 and show the differences expected from random assignment if sleep condition has no effect. The median-based distribution is more discrete and has a somewhat different spread because medians change in jumps and are less sensitive to extreme observations.

5. Use the applet to find and report the *p*-value. To show your work, include a screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) null distribution, (2) statistic and (3) *p*-value.

6. Does this *p*-value indicate strong evidence that sleep deprivation has a harmful effect on improvement score? Explain how you are deciding.

Solution: Yes. The p-value of about 0.025 gives strong evidence that the long-run median improvement score is higher with unrestricted sleep. The evidence is weaker than in the mean-based analysis, but the result still argues against no harmful effect.